

A Privacy-Preserving Workload Logger Using Keyboard and Mouse Dynamics: System Design, Feature Engineering, and a Pilot Study

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February 26, 2026

Abstract

This paper describes a lightweight, privacy-preserving system for collecting interaction dynamics from a standard keyboard and mouse and mapping these signals to subjective workload. The system logs events locally, aggregates features in fixed 5 s windows, and produces a session-level dataset row suitable for machine learning. Subjective workload labels are obtained post-session using a reduced NASA-TLX questionnaire with five dimensions (Mental Demand, Temporal Demand, Performance, Effort, Frustration; excluding Physical Demand). We present the full measurement pipeline, feature set design (including confidence-weighted deletion estimates), and results from a small pilot dataset of seven sessions across two participants and multiple task types. A ridge regression baseline with leave-one-out validation yields negative R^2 across targets, highlighting limitations of dataset size and label noise; we discuss improvements for data collection and modeling that are expected to increase predictive performance.

1 Introduction

Workload and cognitive strain influence performance, learning outcomes, and safety in real-world computer-based tasks. Many workload sensing approaches rely on specialized sensors (e.g., physiological devices), which can limit scalability and adoption. This work aims to enable workload-aware applications using only commodity input signals: keyboard and mouse dynamics.

We contribute:

- A privacy-preserving logger that captures input dynamics without recording typed content.
- A feature extraction pipeline producing window-level and session-level features, including trajectory-based mouse features and confidence-weighted text deletion estimates.
- A pilot dataset and baseline modeling results for five workload dimensions and their mean score.

2 System Overview

2.1 Logging and Privacy

The logger runs locally and records:

- Mouse motion samples (cursor positions), clicks, and scroll events.

- Keyboard press/release timing using privacy-safe key representations (e.g., character keys logged as `CHAR` instead of the actual character).
- Deletion actions (Backspace/Delete) are logged as *estimates* of deleted characters, not actual text.

No raw text is stored. The system writes:

- `input_events.jsonl`: event-level logs plus window aggregates.
- `session_dataset.jsonl`: one row per session with aggregated features and NASA-TLX labels.

2.2 Time Windowing

Signals are aggregated in fixed windows of length $W = 5$ s. For each window, the logger computes movement statistics, click features, trajectory features, and typing features. A summary row is produced for each window, then aggregated into session-level features (mean, standard deviation, min, max across windows).

2.3 Post-Session Subjective Labels

After logging stops, participants complete a reduced NASA-TLX (raw) questionnaire with five dimensions scored 0–100:

1. Mental Demand
2. Temporal Demand
3. Own Performance
4. Effort
5. Frustration Level

The system also computes a session label:

$$\text{TLX Mean} = \frac{1}{5} \sum_{i=1}^5 \text{TLX}_i. \quad (1)$$

(Physical Demand is excluded in this version.)

3 Feature Engineering

3.1 Mouse Features (Window-Level)

For each 5 s window:

- **Distance:** total cursor path length (pixels).
- **Speed:** mean speed computed both time-weighted and sample-weighted; speed standard deviation.
- **Idle time:** inferred from gaps between motion callbacks; idle ratio and active ratio.
- **Clicks:** click count, click rate, click-hold mean and standard deviation.

3.2 Mouse Trajectory Features (Window-Level)

Motion is segmented into trajectories using an inactivity threshold $\tau = 0.15$ s:

- A new trajectory begins after a gap $> \tau$ (treated as idle) or at the start of a window.
- Trajectories are finalized on mouse click or on large time gaps.
- Trajectories outside a duration range $[0.5\text{ s}, 10\text{ s}]$ are discarded.

For each trajectory we compute:

- **Mean trajectory speed:** path length / duration.
- **Efficiency:** straight-line displacement / path length (only if displacement exceeds a minimum threshold).
- **Curvature:** average absolute change in direction between consecutive segments.
- **Direction consistency (“accuracy”):** proportion of consecutive segments that preserve sign of motion direction.
- **Speed–accuracy tradeoff:** $\text{SAT} = \bar{v} \cdot (1 - \overline{\text{accuracy}})$ (computed when accuracy is available).

Window-level trajectory features include counts and the mean/std of trajectory speed, plus mean efficiency, curvature, and direction consistency when available.

3.3 Keyboard Features (Window-Level)

Key dynamics are summarized without storing text content:

- **Typing rate:** # CHAR key-down events per second.
- **Key counts:** total key-down and character key-down counts.
- **Hold times:** mean/std of CHAR key press durations.
- **Pauses:** mean/std of inter-key intervals between consecutive CHAR presses.

3.4 Deletion Estimation with Confidence Weighting

Because the logger does not store actual text, deleted character counts are estimated using heuristics:

- Plain Backspace/Delete $\Rightarrow$ estimate 1 character per action (high confidence).
- Word deletion shortcuts (e.g., Ctrl+Backspace) $\Rightarrow$ estimate based on rolling median word length (medium confidence).
- Selection deletion (keyboard arrows with Shift, mouse drag selection, double click selection) $\Rightarrow$ estimate based on selection size heuristics (confidence depends on method).

Each deletion burst is labeled with a confidence level $\in \{\text{high}, \text{medium}, \text{low}\}$ and mapped to a weight:

$$w(\text{high}) = 1.0, \quad w(\text{medium}) = 0.6, \quad w(\text{low}) = 0.3.$$

The system maintains:

- Total estimated deleted characters.
- Deleted characters split by confidence (high/medium/low).
- A confidence-weighted deleted character count:

$$\text{DelWeighted} = \sum_{b \in \text{bursts}} w(c_b) \cdot \hat{d}_b$$

where $\hat{d}_b$ is the estimated deleted characters for burst b and c_b is its confidence label.

- A window-level deletion confidence score:

$$\text{DelConfScore} = \begin{cases} \frac{\text{DelWeighted}}{\text{DelTotal}} & \text{if DelTotal} > 0 \\ 0 & \text{otherwise.} \end{cases}$$

3.5 Session-Level Features

Each session produces many window rows. These are aggregated into a single training row using:

- Mean and standard deviation of each window feature across windows.
- Minimum and maximum of each window feature across windows.

In the pilot configuration this yields 134 numeric feature columns.

4 Pilot Dataset

4.1 Participants and Tasks

Pilot data contains seven sessions from two participants (“Zafar” and “Dmitriy”) with multiple task types (typing, composing, writing, and question answering). Task metadata (`task_type`, `task_name`) is recorded but excluded from baseline modeling to emphasize input-signal-based prediction.

4.2 Labels

Each session includes five NASA-TLX dimension ratings and their mean. Example label ranges in the pilot dataset span low to high workload (e.g., mental demand from 5 to 80, frustration from 0 to 92), suggesting useful label variation but limited sample size.

5 Modeling and Evaluation

5.1 Baseline Model

We train ridge regression models using a standard preprocessing pipeline:

- Median imputation for missing numeric values.
- Standardization (z-score) of numeric features.
- Ridge regression with hyperparameter α tuned over a log-spaced grid.

5.2 Cross-Validation

Given $n = 7$ sessions, we evaluate using leave-one-out cross-validation (LOO). Hyperparameter tuning uses a nested approach (inner CV) for robustness, though with very small n the variance of estimates remains high.

5.3 Metrics

We report:

- Mean absolute error (MAE)
- Root mean squared error (RMSE)
- Coefficient of determination (R^2)

6 Results

Table 1 reports pilot baseline results for each target. Performance is poor (negative R^2) across all targets, consistent with severe data scarcity, participant effects, and noisy subjective labeling.

Table 1: Pilot results with ridge regression and leave-one-out validation ($n = 7$ sessions).

Target	MAE	RMSE	R^2
TLX Mental Demand	31.129	34.565	-0.613
TLX Temporal Demand	26.086	30.211	-0.425
TLX Performance	24.562	28.368	-0.571
TLX Effort	25.626	28.174	-0.529
TLX Frustration	31.568	40.745	-0.954
TLX Mean (5-dim)	18.560	21.461	-0.572

7 Discussion

7.1 Why R^2 is Negative

Negative R^2 indicates the model performs worse than predicting the mean of the training labels. With only seven samples and 134 features, the model is dominated by variance and regularization. The frequent selection of a large ridge penalty (e.g., $\alpha = 1000$) suggests the best strategy is heavy shrinkage, consistent with weak learnable signal under current sample size.

7.2 Key Limitations

- **Sample size:** $n = 7$ is far below what is typically needed to learn stable mappings from behavioral features to subjective workload.
- **Participant confounds:** only two participants; personal style differences can dominate signals.
- **Task heterogeneity:** mixing tasks (typing vs. essay writing vs. answering) creates distinct interaction patterns; models may require either task conditioning or many more examples per task family.

- **Label noise:** subjective TLX ratings can vary due to mood, interpretation of “performance,” and time of day.

7.3 How to Improve Predictive Performance

Based on the current pipeline, the most impactful changes are:

- **Collect more sessions:** prioritize 50–200+ sessions with repeated tasks per participant and multiple participants.
- **Separate task families or include task metadata:** train per-task-type models or include `task_type/task_name` as categorical inputs.
- **Stabilize labeling:** provide clear TLX instructions (especially for “performance”) and collect ratings immediately after each task segment (rather than a long combined session).
- **Add within-session labels:** short TLX mini-ratings (e.g., every 2–5 minutes) to increase labeled samples without requiring more participants.
- **Feature pruning:** reduce dimensionality by removing redundant min/max stats, using PCA, or using regularized feature selection.

8 Ethical Considerations

The logger is designed to reduce privacy risk by avoiding storage of typed content. Nevertheless, timing and behavioral biometrics can still be sensitive. Recommended safeguards include informed consent, local-only storage by default, encryption at rest, and clear data retention policies.

9 Conclusion

We presented a complete, privacy-preserving keyboard/mouse workload logging pipeline that produces both detailed event logs and session-level ML-ready rows labeled with a reduced NASA-TLX. Pilot ridge regression results on seven sessions show poor generalization, emphasizing the need for substantially more data, clearer task structure, and improved labeling practices. The described system provides a scalable foundation for future, higher-powered studies and model development.

Reproducibility Notes

The implementation outputs:

- `input_events.jsonl`: raw events and window aggregates.
- `session_dataset.jsonl`: one row per session with aggregated features and TLX labels.
- Trained model artifacts and feature importance CSVs from the ML script.

References

- [1] S. G. Hart and L. E. Staveland, “Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research,” in *Human Mental Workload*, P. A. Hancock and N. Meshkati, Eds. North-Holland, 1988.